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Measuring inequality using censored data: a multiple-imputation approach to estimation and inference

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Summary. To measure income inequality with right-censored (top-coded) data, we propose multiple-imputation methods for estimation and inference. Censored observations are multiply imputed using draws from a flexible parametric model fitted to the censored distribution, yielding a partially synthetic data set from which point and variance estimates can be derived using complete-data methods and appropriate combination formulae. The methods are illustrated using US Current Population Survey data and the generalized beta of the second kind distribution as the imputation model. With Current Population Survey internal data, we find few statistically significant differences in income inequality for pairs of years between 1995 and 2004. We also show that using Current Population Survey public use data with cell mean imputations may lead to incorrect inferences. Multiply-imputed public use data provide an intermediate solution.

Keywords: Censored data; Current Population Survey; Generalized beta of the second kind distribution; Income inequality; Multiple imputation; Top coding

1. Introduction

Many data sets that are used to measure income inequality are right censored ('top coded'). To maximize confidentiality and to minimize disclosure risk, data producers do not release files containing complete survey responses. Instead, income values for each income source that are above a source-specific threshold are replaced in public use files by the threshold itself (the 'top code'). Right censoring is a problem for estimation of levels of inequality because it suppresses genuine income dispersion, and it is a problem for estimation of trends in inequality if top code values are not adjusted consistently over time. Top coding also affects estimates

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of standard errors of inequality statistics because variance estimates depend on second- and higher order moments, and their calculation is affected by right censoring. In this paper, we propose a multiple-imputation approach to estimation of and inference about inequality by using top-coded data, illustrating the method with data from the US March Current Population Survey (CPS).

The CPS has been widely used both within and outside government to assess inequality of individual earnings and household income. All CPS data on sources of income are subject to censoring, however, which is a major potential limitation. Top coding in the CPS public use files is common, as we show below. Even the internal CPS data, which are used by the US Census Bureau to produce official income distribution statistics, are also top coded though to a substantially lesser degree (see for example US Census Bureau (2009)). In both types of CPS data, top code values have not been adjusted consistently over time. Burkhauser *et al.* (2008) review CPS topcoding practices and provide references to earlier discussions of top coding.

Although our proposed methods are illustrated with reference to the CPS, they are applicable more widely since the CPS is not the only survey with top-coded data. For example, in the USA, the National Longitudinal Survey of Youth top-codes some of its sources of income as does the Panel Study of Income Dynamics. In the UK, to comply with the Statistics and Registration Service Act of 2007, the Annual Population Survey and the Quarterly Labour Force Survey have introduced top codes on earnings data in public release files. In Germany, the wage data that are available from social insurance administrative registers are right censored at the earnings level corresponding to the upper limit to social insurance contributions.

Without imputation of some kind, point identification of measures of inequality from censored data is impossible in general. Manski (2003) addressed related issues and showed that, with censoring, one may derive an identification region—the lower and upper bounds—of some statistics summarizing censored distributions, notably parameters such as the mean and quantiles (Manski (2003), page 11). Applying these results to internal and public use CPS data, Burkhauser *et al.* (2009) showed that the bounds on the P90/P10 measure of income inequality (the ratio of the 90th percentile to the 10th percentile) vary substantially with the prevalence of top coding, because top-coded observations arise at incomes below the 90th percentile. In addition, there is also the problem that percentile ratio inequality measures are not based on incomes from throughout the income range. This limitation is not shared by inequality measures that are based on all incomes such as Lorenz curves, the Gini coefficient, coefficient of variation and members of the Atkinson and generalized entropy classes of inequality indices—which are all widely used in the academic income distribution literature and official statistics, and reviewed by Cowell (2000). However, Manski's (2003) results do not apply to these measures: with right-censored data, the bounds on the measures are uninformative because the upper bound is the maximum possible value. So, to make further progress with estimation, some form of imputation for censored incomes is necessary.

All previous imputation procedures applied to CPS data that we are aware of have used methods that yield a single imputation for each right-censored value. And only rarely has the sampling variability of the estimates derived from the imputation-augmented data also been estimated in a manner that takes proper account of the right censoring. We propose a multiple-imputation approach to estimating measures of inequality from top-coded data. The approach is closely related to, but was developed independently from, the parametric multiple-imputation method proposed by An and Little (2007) for top-coded data, though we focus on measures of inequality rather than means and regression coefficients as they did.

We show how our approach provides consistent estimates of not only measures of inequality but also their sampling variances, accounting for both stochastic imputation error and sampling

variability. We use the method to analyse recent trends in household income inequality in the USA, exploiting our unprecedented access to internal CPS data.

Throughout the paper, income is defined in a conventional manner. It is household income from all sources excluding capital gains, adjusted for differences in household size by using the square root of household size. It is what US analysts refer to as ‘pre-tax post-transfer’ income. The specific procedures that were followed when constructing the income measure are as discussed by Burkhauser and Larrimore (2009). This definition of income is common in the cross-national comparative income distribution literature (see Atkinson *et al.* (1995)) and studies of US income distribution trends (see Gottschalk and Danziger (2005)). Each individual is attributed with the size-adjusted income of the household to which he or she belongs. Income refers to income for the calendar year preceding the March interview. All references to ‘year’ are to income year rather than survey year. We convert negative and zero household income values each year (which were never more than 1% of cases) to \$1 before our calculations because some indices of inequality are defined only for positive income values. Our samples comprise all individuals in CPS respondent households, excluding individuals in group quarters or in households containing a member of the military. All statistics are calculated using the relevant CPS sampling weights. The weights adjust for non-response associated with observable characteristics, but not unobservables. Sample sizes are large. For example, for 2004, the sample income distribution refers to 207925 individuals in 75660 households.

We compare our multiple-imputation estimates of levels and trends of inequality from the internal CPS data—what we label the Internal-MI series—with two series of estimates derived from public use CPS data which we label Public-CM and Public-MI. The Public-CM series arises when top-coded values are replaced by cell mean imputations derived from internal CPS data. These imputations have been provided by the US Census Bureau for each year since 1995 and are available to all users of public use CPS data. The availability of this series is one reason why we restrict our attention to the period 1995–2004 in this paper. A second reason is that we wish to avoid any potential inconsistencies in the income series arising from the major redesign of the CPS, including the introduction of computer-assisted personal interviewing in survey year 1994 (Ryscavage, 1995). Third, it is well known that US income inequality increased substantially between the mid-1970s and the mid-1990s (see for example Danziger and Gottschalk (1995)), and so we focus on a later period about which there is less unanimity. For the decade starting from the mid-1990s, there is debate about the nature of trends in inequality, but there is agreement that ascertaining trends in the very richest incomes is of particular importance in the USA and a number of other Organization for Economic Co-operation and Development countries (see for example Atkinson and Piketty (2007), Burkhauser *et al.* (2008) and Piketty and Saez (2003)).

The third series we analyse, labelled Public-MI, is also derived from public use CPS data but applies a multiple-imputation approach to estimation and inference that mimics the approach we apply to internal data. Although the internal data provide the best results, researchers can gain access to them only under special conditions, whereas public use data are available to all researchers. It is therefore of interest to explore the extent to which results from the Public-MI series match those from the Internal-MI series.

Using multiply-imputed internal CPS data, we show that the inequality of household income did not change significantly between 1995 and 2004, whether one uses ordinal evaluations that are based on Lorenz curves or cardinal comparisons based on various commonly used inequality indices. We find that the cell-mean-augmented public use data lead to substantial underestimates of levels of inequality in every year, though the trends over time are tracked relatively well. However, the sampling variability of estimates that are derived from the cell-mean-augmented

distributions is also underestimated and, as a result, there is a tendency for trends in inequality over the period to be shown (incorrectly) as statistically significant. Multiply-imputed public use data are shown to provide an intermediate case.

The programs that were used in the analysis can be obtained from

<http://www.blackwellpublishing.com/rss>

2. Right censoring in income data from the March Current Population Survey

In the March CPS, a respondent in each household is asked a series of questions on the sources of income for the household. Starting in 1975, respondents reported income from 11 sources, and since 1987 they have done so for income from 24 sources. High values for each separate source of income are top coded by the Census Bureau; it is not only the highest total household income values that are top coded. See Larrimore *et al.* (2008) for a full list of top code values in the public use and internal CPS data, by source of income.

The prevalence of top coding in total household income is significantly greater than for any specific source of income, since household income is the aggregation of multiple income sources across types of income and household members, and each source may be top coded. As a result, and also because income is measured by using size-adjusted household income rather than nominal household income (see above), top-coded household income values are not necessarily the highest incomes—right-censored observations may occur throughout the income distribution. Hence measuring inequality by using the ratio of the 90th percentile to the 10th percentile (P90/P10) with the goal of minimizing the effect of top coding on estimates of inequality will not be entirely successful: see Burkhauser *et al.* (2009).

The proportion of individuals with top-coded household income in each March CPS from

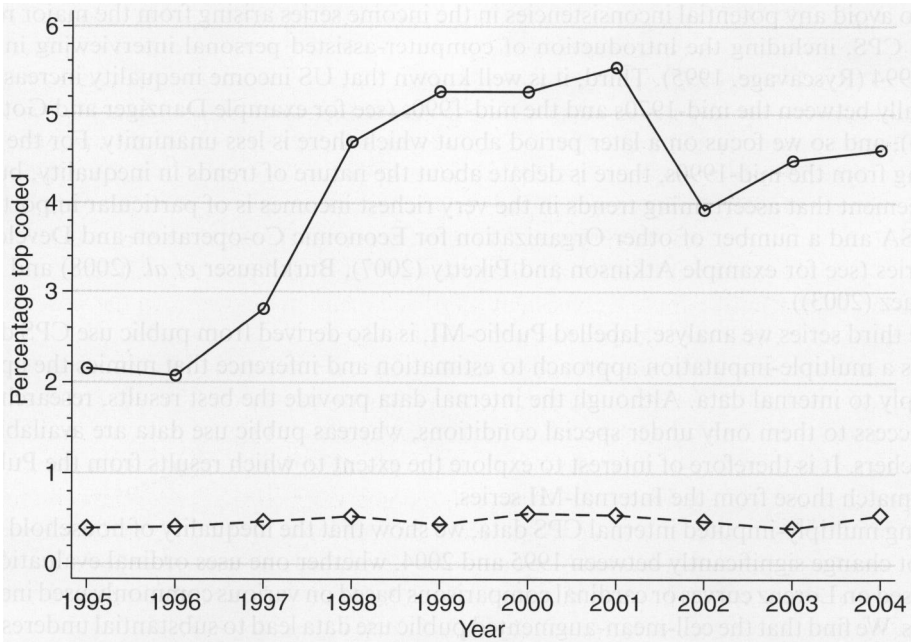


Fig. 1. Percentage of individuals with top-coded household income in the March CPS public use data (—) and internal data (---), by year: authors' calculations from public use and internal March CPS data

1995 through 2004 is shown in Fig. 1. In the public use data, the fraction is substantial, ranging between 2.1% and 5.7%. In the internal data, the proportion is roughly constant and small: only about 0.5%. The much lower prevalence of right censoring in the internal data indicates their substantial value for assessments of inequality compared with public use data. Using internal data rather than the public use data means that incomes are better measured for up to approximately 5.2% ($5.7\% - 0.5\%$) more observations. Nevertheless, censoring remains pervasive in the internal data. The mean size-adjusted household income value for top-coded observations in the internal data is around \$200,000. In the internal data for 1995, the lowest decile of the distribution of top-coded incomes corresponds to the 55th percentile of the distribution for all incomes. For 2000, the lowest decile of the distribution corresponds to the 87th percentile of the overall distribution, and somewhere between these positions in the other years. So, accurate measurement of the degree of inequality needs to account for a non-trivial degree of right censoring, even with CPS internal data. Our multiple-imputation approach to estimation and inference provides this.

3. Single-imputation methods to account for top coding

The two principal imputation approaches to account for top coding in public use CPS data that currently exist are reviewed in this section. The first approach is empirically based, using cell means derived from internal CPS data. Another more *ad hoc* imputation procedure, used for example by Katz and Murphy (1992), Lemieux (2006) and Autor *et al.* (2008), is to replace each top-coded value by a common multiple of the top code. The second approach is model based, assuming that the upper tail of the income distribution has a specific parametric functional form. These approaches yield a single augmented data set for analysis in which each top-coded value is replaced by a single imputed value: they are therefore ‘single-imputation’ methods.

For each year of public use data since income year 1995 (survey year 1996), the Census Bureau has imputed a cell mean value to each top-coded value in the public use data. Before 1995, the public use CPS data for each source of income contained the top code value for each source of income for each observation top coded on that source. These imputations, derived from internal CPS data, are, for each source, equal to the mean source income of all individuals with incomes greater than or equal to the top code, subject to some constraints on minimum cell size. For labour income sources, the means are calculated within cells further divided by race, gender and employment status. The Census Bureau initially provided cell means for wages and salaries, self-employment income and farm income but later extended them to other non-governmental sources of income (1998 and thereafter). Larrimore *et al.* (2008) provide further details of the derivation. They also distribute a consistent set of cell mean imputations to the wider research community that extends the Census Bureau series back to 1975.

These cell mean imputations are a substantial advance for analysis, providing more accurate measures of the incomes of top-coded observations than do the top codes themselves. They have two limitations, however, which lead to underestimation of overall inequality statistics and their sampling variances. First, by construction, all observations within the same cell receive the same imputed value, thereby removing all within-cell income variation. Second, the cell means are derived from internal data which are themselves right censored. This imparts a downward bias to cell mean estimates of top-coded incomes, and hence also a downward bias in estimates of overall inequality and its sampling variance. This bias is of unknown degree since the actual incomes of the censored observations in the internal data are unknown.

The second single-imputation approach was developed before the Census Bureau cell mean series existed. Fichtenbaum and Shahidi (1988), using public use CPS data for 1967–1984,

proposed that the upper tail of the US income distribution for each year be summarized by the one-parameter Pareto distribution. Estimates of the Gini inequality index for each year derived using the Pareto estimates were shown to be between 0.9% and 7.3% greater than corresponding Gini indices estimated ignoring top coding. Essentially the same method was applied by Bishop *et al.* (1994) except that they used unit record public use CPS data rather than grouped data, examined 1985–1989, and compared entire distributions by using Lorenz curves as well as Gini coefficients. Bishop *et al.* (1994) also estimated sampling variances for their inequality statistics and used statistical inference procedures to test differences in inequality. However, these procedures did not take account of the additional variability introduced by the stochastic nature of the imputation process.

In addition to ignoring uncertainty in imputation, the Pareto imputation method has the disadvantage, which is shared with the cell mean imputation approach, that only a single value is imputed to every top-coded observation in the relevant year, so income dispersion is underestimated. Moreover, and as acknowledged by Fichtenbaum and Shahidi (1988) and Bishop *et al.* (1994), the goodness of the Pareto fit to CPS income data is debatable. On this, see for example Angle and Tolbert (1999). This suggests that the use of less restrictive parametric functional forms is productive in this context. In the next section, we make the case for applying the generalized beta distribution of the second kind, GB2 (McDonald, 1984) to CPS data on household incomes, and for using a multiple- rather than single-imputation approach.

4. Multiple-imputation approach to account for top coding

Our multiple-imputation approach consists of five steps, which we outline before discussing in more detail. First we fit an imputation model—which is a parametric functional form that is presumed to describe the income distribution in each year 1995–2004, including right-censored observations. Second, for each observation with a right-censored income, we draw a value from the distribution implied by the fitted model using an appropriate randomization procedure. Third, using the distribution comprising imputed incomes for censored observations and observed incomes for non-censored observations, we estimate our various indices of inequality and associated sampling variances using complete-data methods. Fourth, we repeat the second and third steps 100 times for each year and, finally, we combine the estimates from each of the 100 data sets for each year using the averaging rules proposed by Reiter (2003) for the case of ‘partially synthetic’ data. This accounts for the uncertainty added to estimates by the imputation process as well as for sampling variability. Application of the five-step approach to internal CPS data yields our Internal-MI series of estimates; application to public use CPS data yields our Public-MI series.

Ours is the only study that we are aware of that has applied multiple-imputation methods to right-censored data for the purposes of analysing income inequality. The closest study to ours is An and Little (2007). They fitted log-normal and power-transformed normal distributions to data from the 1995 Chinese household income project and used the estimates to multiply-impute incomes to top-coded observations. Their focus was on estimation of and inference about mean incomes and income regressions for a single year rather than estimates and comparisons of income inequality measures and trends. Gartner and Rässler (2005) used log-normal distributions fitted to German wage data for 1991–2001 to multiply-impute values for top-coded observations. Again the focus was estimation and inference concerning mean incomes and income regressions rather than income inequality.

We assume that the distribution of size-adjusted household income y in each year is described by the four-parameter GB2 distribution, with probability density function

$$f(y) = \frac{ay^{a-1}}{b^ap B(p, q) \{1 + (y/b)^a\}^{p+q}}, \quad y > 0, \quad (1)$$

and cumulative density function

$$F(y) = I\left\{p, q, \frac{(y/b)^a}{1 + (y/b)^a}\right\}, \quad y > 0, \quad (2)$$

where parameters a , b , p and q are each positive. $B(p, q) = \Gamma(p) \Gamma(q) / \Gamma(p + q)$ is the beta function, $\Gamma(\cdot)$ is the gamma function, and $I(p, q, x)$ is the regularized incomplete beta function, which is also known as the incomplete beta ratio. Parameter b is a scale parameter, and a , p and q are each shape parameters. The GB2 distribution is a flexible functional form incorporating many distributions as special cases. For example, the Singh–Maddala (Burr type 12) distribution is the special case of the GB2 distribution when $p = 1$; the Dagum (Burr type 3) distribution is the special case when $q = 1$; and the log-normal distribution is a limiting case. For details, see McDonald (1984) and Kleiber and Kotz (2003). Many studies have shown that the GB2 model fits income distributions extremely well across different times and countries: see *inter alia* McDonald (1984), Bordley *et al.* (1996), Brachmann *et al.* (1996), Bandoorian *et al.* (2003) and Jenkins (2009).

Of particular importance in the current context is the desirable behaviour of the GB2 distribution in its upper tail. Consistent with extreme value theory, the upper and lower tails lie in the domain of attraction of the Fréchet distribution. The upper tail is regularly varying (with variation parameter equal to $-aq$) and it is heavy in that it decays like a power function as income increases, rather than decaying exponentially fast (as for the log-normal distribution, with middle heavy upper tail), or polynomial decreasing (as for Pareto distributions). See Schluter and Trede (2002), appendix A, and Kleiber and Kotz (2003) on regular variation concepts and the upper tail behaviour of GB2 and other distributions.

We estimate the GB2 distribution parameters by maximum likelihood, separately for each year 1995–2004. To ensure that model fit was maximized at the top of the distribution, we fit each GB2 distribution using observations in the richest 70% of the distribution only, making appropriate corrections for left truncation in the maximum likelihood procedure. We chose the 30th percentile as the left truncation point after experiments balancing goodness of fit with ease of maximization. We specify the sample log-likelihood for each year's data as

$$\ln(L) = \sum_{i=1}^N w_i \frac{c_i \ln\{1 - F(y_i)\} + (1 - c_i) \ln\{f(y_i)\}}{1 - F(z)} \quad (3)$$

where $i = 1, \dots, N$ indexes each individual sample observation, w_i is the sample weight for i and $c_i = 1$ if i is an observation with a right-censored household income value, and $c_i = 0$ otherwise. The denominator of the expression adjusts for left truncation: z is the level of income corresponding to the left truncation point. For maximization, we use the modified Newton–Raphson procedure implemented in Stata's `m1` command (StataCorp, 2007), with the parameter covariance matrix estimates based on the negative inverse Hessian. Convergence was achieved easily within several iterations. For brevity, we do not report estimates for each year but they are available from the authors on request.

For the internal data, the model fit varied slightly across years but was generally excellent. This is demonstrated first by the precision of the parameter estimates. For example, the smallest t -ratio for any parameter estimate (always for p) was greater than 7 and was typically at least two or three times larger for a and for q , and in between for b . Wald tests of parameter values suggested that we could easily reject restricted models such as the Singh–Maddala or Dagum

distributions in favour of the GB2 distribution. Excellent goodness of fit to the internal data is also demonstrated by probability plots for each year shown in Fig. 2 of Jenkins *et al.* (2009). These are plots of the cumulative probabilities of income expected given the estimated GB2 parameters against the cumulative probabilities of income observed in the data. Each chart is based on the richest 70% of each distribution for each year, for the reasons explained earlier. Excellent goodness of fit is demonstrated by the fact that every plot lies extremely close to a 45°-ray from the origin, and much closer than is typically observed in plots of this type. There is also no obvious change in the nature of the plots around the neighbourhood of the very richest observations. Such smoothness increases our confidence in the use of the GB2 distribution for imputing incomes to right-censored observations in the internal CPS data.

We also fit GB2 models separately to public use data for each year, using procedures that mimicked those for the internal data. Given the greater prevalence of right censoring, the model goodness of fit was not quite as good as for the internal data, but very good nonetheless. This is illustrated by the probability plots shown by Jenkins *et al.* (2009), appendix Fig. A.

The second stage of our multiple-imputation approach uses the GB2 estimates to derive imputed values for top-coded observations for each year by using the inverse transform sampling method. Given the fitted GB2 cumulative density function $\hat{F}(y)$, the corresponding cumulative density function for top-coded observation i is, using standard formulae for truncated distributions,

$$\hat{G}(y_i) = \frac{\hat{F}(y_i) - \hat{F}(t_i)}{1 - \hat{F}(t_i)} \quad (4)$$

where t_i is the top code for i , and y_i is the ‘true’ value for that observation (which we are unable to observe). Letting $u_i = \hat{G}(y_i)$, and inverting the expression for the income distribution among top-coded observations, we have

$$y_i = \hat{F}^{-1}[u_i\{1 - \hat{F}(t_i)\} + \hat{F}(t_i)]. \quad (5)$$

A value of y_i for each top-coded observation is generated by substituting into this expression a value of u_i that is equal to a random draw from a standard uniform distribution. An alternative approach to taking draws (which was suggested by a referee) would have been to use Halton or Hammersley sequences. Because we use a household level definition of income, all individuals within the same household are imputed with the same value.

The combination of the observed incomes for non-top-coded observations with the imputed incomes for top-coded observations produces a partially synthetic data set for each year to which we can apply complete-data methods to estimate our inequality statistics of interest. (Fully synthetic data consist of entirely multiply-imputed data.) Repetition of the process $m > 1$ times produces m partially synthetic data sets for each year and, correspondingly, m sets of inequality estimates for each year which we combine in a manner that is discussed shortly. Note that the observations without censored data are common across each of the m partially synthetic data sets. Clear-cut rules for the choice of m do not exist, but the number used is often relatively small (10 or fewer). In the next section, we report estimates based on $m = 100$. In preliminary research, we used $m = 20$ and derived similar conclusions to those reported here.

For inference from our multiply-imputed partially synthetic data sets, we use the combination formulae derived by Reiter (2003), and also used by An and Little (2007), as follows.

Suppose that inference is required about some scalar Q , where Q is a measure of inequality such as the Gini index, and index the partially synthetic data sets by $j = 1, \dots, m$. Denote the point estimator of Q from partially synthetic data set j by q_j and the estimator of its variance by v_j . Reiter (2003) showed that one should estimate Q by using the mean of the point estimators

$$\bar{q}_m = \sum_{j=1}^m q_j / m \quad (6)$$

and use

$$T_p = b_m / m + \bar{v}_m \quad (7)$$

to estimate the variance of $\bar{q}_m$, where

$$b_m = \sum_{i=1}^m (q_i - \bar{q}_m)^2 / (m - 1), \quad (8)$$

$$\bar{v}_m = \sum_{j=1}^m v_j / m.$$

Thus, the multiple-imputation point estimate is the simple average of the point estimates that are derived using complete-data methods from each of the m partially synthetic data sets. The variance of this estimate is the average of the sampling variances plus a term reflecting the finite number of imputations, m . T_p differs from Rubin's (1987) rule for the combination of estimates in the fully synthetic data case, in which case $T_p = b_m + b_m / m + \bar{v}_m$. The expression for the fully synthetic data case includes additional variability (the term b_m) to average over the response mechanism (Rubin, 1987). By contrast, '[t]his additional averaging is unnecessary in partially synthetic data settings since the selection mechanism ... is not treated as stochastic' (Reiter (2003), page 5). The selection mechanism in our case refers to the choice of top codes by the Census Bureau for the CPS. For large sample sizes, inference concerning Q can be based on t -distributions with degrees of freedom $v_p = (m - 1)(1 + r_m^{-1})^2$ where $r_m = m^{-1} b_m / \bar{v}_m$. Because our sample sizes are large and m is large, v_p is very large also, so the t -distribution is approximated very well by a normal distribution and that is what we use for inference.

Because cardinal indices of inequality differ in their sensitivities to differences in income in different ranges of the income distribution (Atkinson, 1970), we estimate inequality indices that reflect this feature in a systematic way. Specifically, we consider the mean log-deviation, MLD, Theil index and half the coefficient of variation squared, $CV^2/2$, plus the Gini coefficient. The first three indices belong to the one-parameter generalized entropy class $GE(\alpha)$ with parameter $\alpha = 0, 1, 2$ respectively and range from being bottom sensitive (MLD) to being sensitive to differences in income at the top of the distribution ($CV^2/2$). The commonly used Gini coefficient is a middle sensitive inequality index. For a review of inequality index properties and index formulae, see Cowell (2000). We computed distribution-free variance estimates for the inequality indices according to formulae provided by Biewen and Jenkins (2006) for generalized entropy indices and by Kovačević and Binder (1997) for the Gini index. In both cases, we account for the clustering of individuals in households using a conventional sandwich estimator. The computations were undertaken using the Stata modules provided by Biewen and Jenkins (2005) and Jenkins (2006).

We checked whether our estimates of inequality trends were robust to the choice of inequality measure by investigating whether or not Lorenz curves of income distributions for pairs of years do not intersect. The statistics of relevance for each year are the cumulative shares of income for different income groups for the sample ranked in ascending order of income (the Lorenz ordinates), and their sampling variances. If there is Lorenz dominance, then there is a unanimous ordering of income distributions according to all standard indices of inequality (Foster, 1985). These include the four indices that were mentioned in the previous paragraph. The computations of the distribution-free variance formulae provided by Kovačević and Binder

(1997) were undertaken using the Stata module provided by Jenkins (2006). Again we account for the clustering of individuals in households using a conventional sandwich estimator.

Reiter (2003) provided a Bayesian derivation for his data combination inference formulae, and also the two conditions under which the inferences are valid from a frequentist perspective: that the analyst uses randomization valid estimators and that the synthetic data generation methods are proper in a sense similar to Rubin (1987). Of necessity we argue that these conditions are satisfied in our case; it is impossible to test their validity since access to the actual values for top-coded observations in the internal CPS data is not allowed. Reiter developed his combination rules under the assumption that the producer of the multiply-imputed partially synthetic data sets had used imputation methods that satisfy the conditions. Our imputation procedures are randomization based though not fully Bayesian since we did not draw from the posterior predictive distribution of the GB2 parameters in each year. Because of the complexity of implementing this procedure in our context, we drew from the full data posterior distribution, treating the GB2 parameters as known, and appeal to the excellent fit of our GB2 models. An and Little (2007) employed the same procedure as ours when they derived multiple imputations for data that were assumed to follow a power-transformed normal distribution.

5. Estimates of US income inequality, 1995–2004

We first discuss the results from our analysis of Lorenz dominance and then the indices of inequality. Throughout, tests for statistically significant differences are based on changes between distributions at least 2 years apart. We do not test for differences between adjacent years because of the rotation group structure of the CPS (about half of the sample is the same in the March CPS for consecutive years): the test procedures that we use are predicated on having independent samples in each year. The focus is on the Internal-MI series, but supplemented by discussion of how the estimates from the Public-CM and Public-MI series compare. Although the latter two series underestimate inequality measures and associated sampling variances (as explained earlier), it is of interest to know whether these features lead to erroneous conclusions about differences in inequality.

Detailed results from the Lorenz dominance analysis of the Internal-MI data for the beginning, middle and end of the period (1995, 2000 and 2004) are reported in Table 1. The details for the other years are not shown for brevity. Shown are the estimated Lorenz ordinates (cumulative income shares) at each successive 20th of the distributions, together with the estimated standard errors (in parentheses) derived using the methods discussed in the previous section. The rightmost three columns report distribution-free Lorenz dominance test statistics for pairwise comparisons between the three years. For a pairwise comparison between year A and year B, and income group $k = 1, \dots, 19$, each test statistic Δ_k is

$$\Delta_k = (\hat{L}_k^A - \hat{L}_k^B) / \sqrt{(\hat{V}_k^A + \hat{V}_k^B)}, \quad (9)$$

where $\hat{L}_k^A$ is the estimate of the k th Lorenz ordinate and $\hat{V}_k^A$ is the estimate of its variance, and correspondingly for year B. Testing of the null hypothesis of dominance uses the multiple comparison union–intersection method of Bishop *et al.* (1991a,b) and Bishop *et al.* (1994). Tests are based on a 5% level of significance and take account of the fact that each dominance test is based on 19 simultaneous tests. The critical value is therefore obtained from the Student maximum modulus distribution (Beach and Richmond, 1985): $\text{SMM}(19, \infty) = 3.01$.

There are four possible outcomes from the tests associated with the comparison of years A and B, as Bishop and colleagues explained. First, there may be no statistically significant

Table 1. Lorenz ordinates, standard errors and test statistics for pairwise Lorenz comparisons (Internal-MI series)†

Cumulative population share	Estimates for the following years:			SMM test statistics (Δ_k)		
	1995	2000	2004	2004 versus 1995	2004 versus 2000	2000 versus 1995
0.05	0.0040 (0.0063)	0.0040 (0.0081)	0.0031 (0.0057)	−0.1104	−0.0938	−0.0006
0.10	0.0134 (0.0063)	0.0137 (0.0081)	0.0122 (0.0057)	−0.1506	−0.1511	0.0215
0.15	0.0266 (0.0062)	0.0268 (0.0081)	0.0249 (0.0057)	−0.1983	−0.1888	0.0187
0.20	0.0433 (0.0062)	0.0432 (0.0080)	0.0411 (0.0056)	−0.2615	−0.2200	−0.0031
0.25	0.0633 (0.0062)	0.0629 (0.0080)	0.0603 (0.0056)	−0.3550	−0.2660	−0.0355
0.30	0.0866 (0.0061)	0.0857 (0.0080)	0.0826 (0.0056)	−0.4740	−0.3179	−0.0828
0.35	0.1132 (0.0061)	0.1118 (0.0079)	0.1082 (0.0055)	−0.6050	−0.3673	−0.1423
0.40	0.1432 (0.0060)	0.1407 (0.0079)	0.1371 (0.0055)	−0.7391	−0.3677	−0.2513
0.45	0.1764 (0.0060)	0.1730 (0.0078)	0.1693 (0.0055)	−0.8784	−0.3865	−0.3465
0.50	0.2133 (0.0059)	0.2088 (0.0078)	0.2051 (0.0054)	−1.0172	−0.3920	−0.4536
0.55	0.2538 (0.0058)	0.2482 (0.0077)	0.2445 (0.0054)	−1.1657	−0.3918	−0.5736
0.60	0.2982 (0.0058)	0.2915 (0.0077)	0.2880 (0.0053)	−1.2952	−0.3737	−0.6953
0.65	0.3467 (0.0057)	0.3389 (0.0076)	0.3358 (0.0053)	−1.4017	−0.3282	−0.8237
0.70	0.4000 (0.0056)	0.3907 (0.0076)	0.3886 (0.0052)	−1.4915	−0.2338	−0.9858
0.75	0.4587 (0.0055)	0.4479 (0.0075)	0.4474 (0.0051)	−1.4914	−0.0596	−1.1523
0.80	0.5240 (0.0055)	0.5116 (0.0074)	0.5119 (0.0051)	−1.6227	0.0363	−1.3484
0.85	0.5975 (0.0053)	0.5834 (0.0073)	0.5856 (0.0050)	−1.6316	0.2415	−1.5526
0.90	0.6824 (0.0052)	0.6673 (0.0072)	0.6711 (0.0049)	−1.5703	0.4469	−1.7013
0.95	0.7863 (0.0050)	0.7713 (0.0070)	0.7765 (0.0047)	−1.4111	0.6247	−1.7446

†Standard errors are in parentheses. Source: authors' calculations from internal CPS data.

difference between any pair of Lorenz ordinates ($|\Delta_k| \leq 3.01$ for all k), in which case A and B cannot be differentiated. Second, if there are positive and statistically significant differences in ordinates and no negative and statistically significant differences, then year A Lorenz dominates B: inequality is lower according to all standard inequality indices ($\Delta_k > 3.01$ for some k and $|\Delta_l| \leq 3.01$ for $l \neq k$). Third, the reverse is the case if there are negative and statistically significant differences in ordinates and no positive and statistically significant differences ($\Delta_k < -3.01$ for some k and $|\Delta_l| \leq 3.01$ for $l \neq k$). Fourth, if there are both negative and positive differences that are statistically significant ($\Delta_k > 3.01$ for some k and $\Delta_l < -3.01$ for some $l \neq k$), the Lorenz

curves cross and the null hypothesis of Lorenz dominance is rejected. For a review of this and other approaches to testing dominance, see Davidson and Duclos (2000). Some more recently proposed tests such as those of Barrett and Donald (2003) are not widely used and their more computationally intensive nature makes it infeasible to apply them to CPS data in the restricted access Census Research Data Center environment.

Table 1 suggests that, according to the point estimates of the ordinates, the Lorenz curve moved slightly outwards between 1995 and 2004 (indicating greater inequality). However, all the test statistics Δ_k are smaller than 3.01, and so we cannot differentiate between the pairs of Lorenz curves in question. For the two subperiods, the pattern of change in the ordinates is more complex—there are both positive and negative changes in the ordinates—but the outcome is the same as before. Indeed, all 36 pairwise comparisons undertaken (using all pairs of years 1995–2004 excluding adjacent years) fall under the heading of the first case. Thus, according to Lorenz dominance tests applied to multiply-imputed internal data, no significant change in inequality according to all Lorenz consistent inequality measures could be detected over the period 1995–2004.

We repeated the 36 pairwise Lorenz dominance tests using the Public-CM data and found the same result, but with one difference. The null hypothesis of Lorenz dominance could not be rejected for comparisons involving 1999—inequality for this year was apparently lower than for any of the other years that were considered. For example, in the comparison between 1999 and 2004, $\Delta_k > 3.01$ for $k = 12$ –19 inclusively, and $0 < \Delta_k < 3.01$ otherwise, i.e. cumulative income shares were significantly lower in the top third of the income distribution in 2004 compared with 1999. Apart from the results for 1999, there is consistency between the conclusions derived from the Internal-MI and Public-CM series. We discuss why 1999 is different below.

Lorenz dominance tests based on the Public-MI data led to slightly different results. There was less inequality in 1999 than in 1995 and in 1996. But, besides the results for this year, there is broad consistency between these estimates and those from the Internal-MI series.

Stronger conclusions about differences in inequality may be derived by moving from Lorenz dominance tests—concerned with differences according to *all* Lorenz consistent measures of inequality—to comparisons of specific Lorenz consistent indices. Estimates of indices of inequality and test statistics for pairwise comparisons for selected years are shown in Table 2 for the Internal-MI series. The test statistics are for pairwise difference-in-means *t*-tests, and so the relevant critical value using a 5% level of significance is approximately 1.96. We find that the estimate of each index increased between 1995 and 2000 and between 2000 and 2004. The estimated increase between 1995 and 2004 is largest for the GE(2) and GE(0) indices (25% and 18% respectively), and smallest for the Gini and GE(1) indices (3% and 9% respectively). However it is only for the Gini and GE(0) indices that the increases are statistically different from 0. Few subperiod increases are statistically significant either—the exceptions mainly concern GE(0). Observe that the standard errors for the GE(2) index are relatively large, and we comment further on this below.

The complete set of tests for pairwise inequality index differences is summarized in Table 3 for all three series. For each year, index of inequality and source of data series, the cell entry shows the year(s) for which there is a statistically significant difference in inequality between the comparison year (first column) and the year(s) shown. A blank cell means that no comparison between that year and any other year is statistically significant. Table 3 points to several findings about differences in inequality and about consistency in results across series.

First, according to the gold standard of the Internal-MI series, few differences in inequality are significantly different from 0. Where they are statistically significant, they typically refer to differences between the beginning of the period and the end of the period. The greatest

Table 2. Inequality indices, standard errors and test statistics for pairwise comparisons (Internal-MI series)†

Index	Estimates for the following years:			Test statistics		
	1995	2000	2004	2004 versus 1995	2004 versus 2000	2000 versus 1995
Gini	0.4311 (0.0036)	0.4427 (0.0049)	0.4453 (0.0033)	2.8944	0.4391	1.9019
GE(0)	0.4017 (0.0070)	0.4298 (0.0094)	0.4756 (0.0073)	7.2817	3.8536	2.3992
GE(1)	0.3731 (0.0153)	0.4166 (0.0265)	0.4066 (0.0166)	1.4836	−0.3176	1.4185
GE(2)	0.9572 (0.2326)	1.4961 (0.9855)	1.1937 (0.5001)	0.4288	−0.2737	0.5322

†Standard errors are in parentheses. Source: authors’ calculations from internal CPS data.

number of statistically significant differences refers to the GE(0) index. Also, none of the tests for differences in the top-sensitive index GE(2) based on the Internal-MI series is statistically significant.

Second, we find more statistically significant differences using the Public-CM series than the Internal-MI series. For some reference years when comparisons based on the Internal-MI series yield no statistically significant differences at all, there are statistically significant differences according to the Public-CM series. Consider, for example, the comparisons for each of the reference years 1997, 1999 and 2000 for the Gini and GE(1) indices, and all comparisons for GE(2) except those involving 1998. And, whenever there is any statistically significant difference concerning a year A and a year B according to the Internal-MI series, there are often statistically significant differences between year A and additional years as well according to the other two series. For example, according to the Internal-MI series, there was a statistically significant difference between the Gini indices for 2004 and 1995. According to the Public-CM series, the Gini index for 2004 differed from the estimates for 1995, 1996, 1998, 1999 and 2000. In only a few instances was there a statistically significant difference according to the Internal-MI series but not the Public-CM series, notably the GE(0) comparisons for 2001 with 1997 and 1998, and 2002 with 1999.

The explanation for these findings is that the suppression of genuine within-cell income dispersion associated with cell mean imputation leads to underestimation of not only indices of inequality but also of their sampling variances. There is therefore a tendency for estimated trends in inequality based on the Public-CM series to be judged statistically significant when they are not.

The differences between the three series are highlighted by Fig. 2. This shows the estimates for each of the four indices of inequality, year by year, together with the associated 95% confidence intervals. To ensure that the lower bound of the confidence interval for GE(2) is positive, the interval for this index was calculated in the logarithmic metric and then transformed back to the natural metric.

Unsurprisingly, estimates of levels of inequality in each year are greater for the Internal-MI series than for the Public-CM series. However, both series point to similar trends over the period:

Table 3. Pairwise comparisons of inequality indices, by index, year and estimate series†

Year	Gini			GE(0)			GE(1)			GE(2)		
	Internal-MI	Public-CM	Public-MI	Internal-MI	Public-CM	Public-MI	Internal-MI	Public-CM	Public-MI	Internal-MI	Public-CM	Public-MI
1997		1995	1995	1995	1995	1995		1995	1995		1995	
1998			1995, 1996	1995, 1996	1995, 1996	1995, 1996						
1999		1995–1997	1995–1997		1997	1995, 1996		1995–1997	1995, 1996		1995–1997	
2000			1995, 1996	1995	1995	1995, 1996			1995			
2001	1995, 1996, 1999	1995, 1996, 1998, 1999	1995, 1996, 1999	1995	1995	1995, 1996	1995	1995, 1998, 1999	1995		1997	1999
2002		1995, 1999	1995, 1996, 1999	1995, 1996, 1999	1995, 1996, 1999, 2000	1995, 1996		1995, 1999	1995		1995, 1999, 2000	1999
2003	1995	1995, 1998, 1999	1995, 1996, 2001	1995–2000	1995–2000	1995–2001		1995, 1999	1995, 1996			
2004	1995	1995, 1996, 1998–2000	1995, 1996	1995–2000, 2002	1995–2002	1995–2002		1995, 1998, 1999, 2000	1995		1995, 1998, 1999	2000

†For each year, index of inequality and source of data series, the cell entry shows the year(s) for which there was a statistically significant difference in inequality between that year and the year(s) shown. A blank cell means no comparison between that year and any other year was statistically significant. Comparisons undertaken for every pair of years 1995–2004, adjacent years excepted; authors' calculations from internal and public use CPS data.

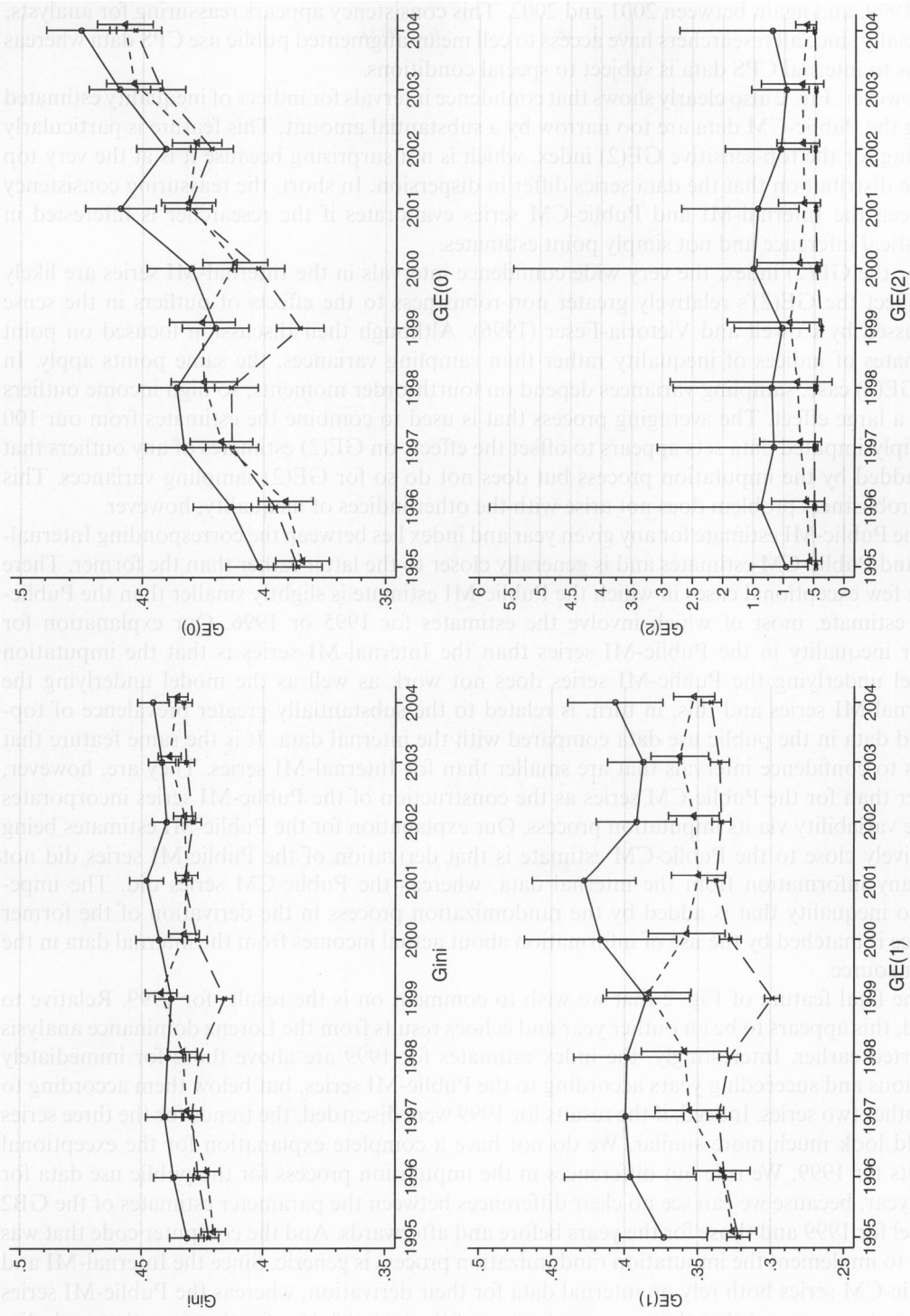


Fig. 2. Inequality indices with 95% confidence intervals, 1995–2004 (—, Internal-MI series; ---, Public-MI series; ·····, Public-MI series)

they suggest that levels of inequality fell slightly at the end of the 1990s, especially between 1998 and 1999, and again between 2001 and 2002. This consistency appears reassuring for analysts, especially since all researchers have access to cell mean-augmented public use CPS data whereas access to internal CPS data is subject to special conditions.

However, Fig. 2 also clearly shows that confidence intervals for indices of inequality estimated using the Public-CM data are too narrow by a substantial amount. This feature is particularly striking for the top-sensitive GE(2) index, which is not surprising because it is at the very top of the distribution that the data series differ in dispersion. In short, the reassuring consistency between the Internal-MI and Public-CM series evaporates if the researcher is interested in statistical inference and not simply point estimates.

For the GE(2) index, the very wide confidence intervals in the Internal-MI series are likely to reflect the GE(2)'s relatively greater non-robustness to the effects of outliers in the sense discussed by Cowell and Victoria-Feser (1996). Although their discussion focused on point estimates of indices of inequality rather than sampling variances, the same points apply. In the GE(2) case, sampling variances depend on fourth-order moments, so high income outliers have a large effect. The averaging process that is used to combine the estimates from our 100 multiply-imputed data sets appears to offset the effects on GE(2) estimates of any outliers that are added by the imputation process but does not do so for GE(2) sampling variances. This non-robustness problem does not arise with the other indices of inequality, however.

The Public-MI estimate for any given year and index lies between the corresponding Internal-MI and Public-CM estimates and is generally closer to the latter rather than the former. There are a few exceptional cases in which the Public-MI estimate is slightly smaller than the Public-CM estimate, most of which involve the estimates for 1995 or 1996. Our explanation for lower inequality in the Public-MI series than the Internal-MI series is that the imputation model underlying the Public-MI series does not work as well as the model underlying the Internal-MI series and this, in turn, is related to the substantially greater prevalence of top-coded data in the public use data compared with the internal data. It is the same feature that leads to confidence intervals that are smaller than for Internal-MI series. They are, however, larger than for the Public-CM series as the construction of the Public-MI series incorporates more variability via its imputation process. Our explanation for the Public-MI estimates being relatively close to the Public-CM estimate is that derivation of the Public-MI series did not use any information from the internal data, whereas the Public-CM series did. The impetus to inequality that is added by the randomization process in the derivation of the former source is matched by the use of information about actual incomes from the internal data in the latter source.

The final feature of Fig. 2 that we wish to comment on is the results for 1999. Relative to trend, this appears to be an outlier year and echoes results from the Lorenz dominance analysis reported earlier. Interestingly, the index estimates for 1999 are above those for immediately previous and succeeding years according to the Public-MI series, but below them according to the other two series. Indeed, if the results for 1999 were discarded, the trends for the three series would look much more similar. We do not have a complete explanation for the exceptional results for 1999. We rule out differences in the imputation process for the public use data for that year, because we can see no clear differences between the parameter estimates of the GB2 model for 1999 and those for the years before and afterwards. And the computer code that was used to implement the imputation randomization process is generic. Since the Internal-MI and Public-CM series both rely on internal data for their derivation, whereas the Public-MI series does not, we suspect that there is some feature of the internal data for that year that underlies the pattern.

6. Discussion

We have demonstrated how a multiple-imputation approach may be used to estimate levels of inequality and trends from right-censored income data. With a suitable imputation model, researchers may impute values to top-coded observations, thereby creating multiple partially synthetic data sets to be analysed by using complete-data methods. Estimates combined using straightforward formulae can be used for statistical inference.

Applying the multiple-imputation approach to internal data from the CPS, we have shown that no clear-cut conclusions about the changes in income inequality over the period between 1995 and 2004 can be drawn. According to Lorenz dominance analysis, there was no significant change in inequality. For some specific indices, such as the Gini or the GE(0) index, there was a statistically significant increase in inequality between 1995 and 2004; for more top-sensitive indices such as GE(1) and GE(2), changes did not differ significantly from 0.

Our analysis enables assessment of the public use CPS data augmented by the Census Bureau's cell mean imputations for top-coded observations in the context of estimation of inequality in income and its trends. Taking the estimates from the multiply-imputed internal data as the gold standard shows that the cell mean-augmented data track trends in indices of inequality over the decade since 1995 reasonably well, though levels of inequality are—unsurprisingly—underestimated. However, we have also shown that suppression of income dispersion within cells, combined with use of right-censored CPS internal data to construct cell means, also has effects on variance estimates. Compared with their multiply-imputed internal data counterparts, they are underestimated, leading to confidence intervals that are too narrow and a tendency to find statistically significant differences in inequality incorrectly. Put another way, cell mean imputations for top-coded observations may do an excellent job of helping to estimate mean incomes, but their very nature makes them less suitable for estimation and inference concerning statistics based on higher order moments.

Although few researchers would find it practical to go through the procedures that are required to access the internal CPS data, and to undertake the research using the data within a US Census Bureau Data Center, we have shown that there is a feasible alternative that works reasonably well, i.e. our comparisons of the three series show that a multiple-imputation approach applied to top-coded public use CPS data can yield results about inequality in income that in several senses lie between those derived using multiple imputation applied to internal data and those derived using cell mean-augmented public use data. The Public-MI approach takes account of income dispersion at the top of the distribution and also takes account of the variability of estimates. Because the public use data are, by definition, in the public domain, it would also be easier for researchers to build more sophisticated imputation models and to improve the quality of estimates that are derived. These models might allow for subgroup differences, for instance allowing for covariates in the estimation of a parametric model or also incorporating the information that is available from the cell mean imputations. As argued in Section 1, the multiple-imputation methods that we propose may be applied to a number of data sets other than the CPS. The methods might also be applied to the case of interval censoring, i.e. when income data are provided in grouped form. Imputing a value equal to the midpoint of an interval is a commonly used single-imputation approach (see for example Kahn (1998)); multiple imputation may be a fruitful alternative.

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